04.12.2020

Exerciții pregătitoare pt L.P. (An 1 FI) + (An 1 ID)

$$\sum_{n=1}^{\infty} \frac{1}{(4n-3)(4n+1)} \; ; \; a_n = \frac{1}{(4n-3)(4n+1)} > 0$$

$$\sum_{n=1}^{\infty} \frac{1}{(2n-1)(2n+1)} \; ; \; a_n = \frac{1}{(2n-1)(2n+1)} > 0$$ → serie cu termeni > 0.

Criteriul raportului

$$\lim_{n\to\infty} \frac{a_{n+1}}{a_n} = \lim_{\infty} \frac{\frac{1}{(2n+1)(2n+3)}}{\frac{1}{(2n-1)(2n+1)}} = \lim_{\infty} \frac{2n-1}{2n+3} = \lim \frac{n(2-\frac{1}{n})}{n(2+\frac{3}{n})}$$

$\lim \frac{a_{n+1}}{a_n} = 1$ → inconcludent

Raabe – Dukamel : $\lim_{n\to\infty} n\left(\frac{a_n}{a_{n+1}} - 1\right) = \lim n\left(\frac{2n+3}{2n-1} - 1\right) =$

$= \lim n \cdot \frac{2n+3-2n+1}{2n-1} = \lim \frac{4n}{2n-1} = \lim \frac{4n}{n(2-\frac{1}{n})} = 2 > 1$

⇒ serie convergentă.

$$\sum_{n=1}^{\infty} a_n = \lim_{n\to\infty} S_n = \lim_{n\to\infty} (a_1 + a_2 + \dots + a_n) =$$

$$= \lim_{\infty} \left(\frac{1}{1\cdot 3} + \frac{1}{3\cdot 5} + \dots + \frac{1}{(2n-1)(2n+1)}\right)$$

$$\frac{1}{(2n-1)(2n+1)} = \frac{A}{2n-1} + \frac{B}{2n+1} = \frac{A(2n+1) + B(2n-1)}{(2n-1)(2n+1)}$$

$\Rightarrow (2n+1)\cdot A + (2n-1)\cdot B = 1, \; (\forall) n \in \mathbb{N}.$

$n = -\frac{1}{2} \Rightarrow -2B = 1 \Rightarrow B = -\frac{1}{2}$

$n = \frac{1}{2} \Rightarrow 2A = 1 \Rightarrow A = \frac{1}{2}$

$\Rightarrow a_n = \frac{1}{2}\left(\frac{1}{2n-1} - \frac{1}{2n+1}\right)$

$n=1: a_1 = \frac{1}{2}\left(\frac{1}{1} - \cancel{\frac{1}{3}}\right)$

$n=2: a_2 = \frac{1}{2}\left(\cancel{\frac{1}{3}} - \cancel{\frac{1}{5}}\right)$

$n=3: a_3 = \frac{1}{2}\left(\cancel{\frac{1}{5}} - \cancel{\frac{1}{7}}\right)$

$\dots$

$n-1: a_{n-1} = \frac{1}{2}\left(\cancel{\frac{1}{2n-3}} - \cancel{\frac{1}{2n-1}}\right)$

$n: a_n = \frac{1}{2}\left(\cancel{\frac{1}{2n-1}} - \frac{1}{2n+1}\right)$

(+) $\Rightarrow S_n = a_1 + a_2 + \dots + a_n =$

$= \frac{1}{2}\left(1 - \frac{1}{2n+1}\right)$

$S = \lim_{n\to\infty} S_n = \frac{1}{2}.$

Student:
Grupa:

$$\lim_{n\to\infty} \frac{1+\sqrt{2}+\sqrt{3}+\dots+\sqrt{n}}{n\sqrt{n}} = \lim_{n\to\infty} \frac{x_n}{y_n}$$

$$\lim_{n\to\infty} \frac{x_{n+1}-x_n}{y_{n+1}-y_n} = \lim_{n\to\infty} \frac{\cancel{1+\sqrt{2}+\dots+\sqrt{n}}+\sqrt{n+1}-\cancel{1-\sqrt{2}-\dots-\sqrt{n}}}{(n+1)\sqrt{n+1}-n\sqrt{n}} =$$

$$= \lim_{\infty} \frac{\sqrt{n+1}}{(n+1)\sqrt{n+1}-n\sqrt{n}} \overset{\infty}{\underset{\infty-\infty}{=}} \lim \frac{\sqrt{n+1}\left((n+1)\sqrt{n+1}+n\sqrt{n}\right)}{(n+1)^2\cdot(n+1)-n^2\cdot n} =$$

$$(a+b)(a-b) = a^2-b^2$$

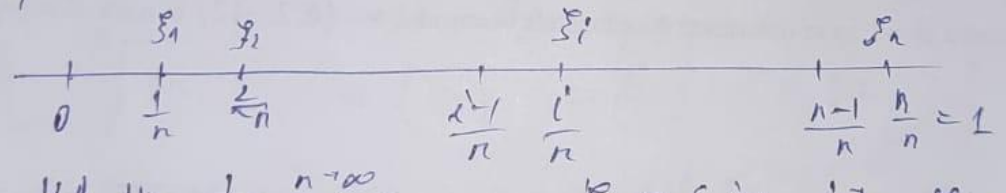
$$= \lim_{\infty} \frac{(n+1)^2+n\cdot\sqrt{n^2+n}}{(n+1)^3-n^3} = \lim_{\infty} \frac{n^2\left[\left(1+\frac{1}{n}\right)^2+\sqrt{1+\frac{1}{n}}\right]}{\cancel{n^3}+3n^2+3n+1-\cancel{n^3}} =$$

$$= \lim_{\infty} \frac{\cancel{n^2}\left[\left(1+\frac{1}{n}\right)^2+\sqrt{1+\frac{1}{n}}\right]}{\cancel{n^2}\left(3+\frac{3}{n}+\frac{1}{n^2}\right)} = \frac{2}{3} \Rightarrow \boxed{\lim_{n\to\infty} \frac{x_n}{y_n} = \frac{2}{3}}$$

$$\lim_{n\to\infty} \frac{1}{n}\left[\sqrt{\frac{1}{n}}+\sqrt{\frac{2}{n}}+\sqrt{\frac{3}{n}}+\dots+\sqrt{\frac{n}{n}}\right]$$

$f(x)=\sqrt{x}$; $\lim_{n\to\infty}\sqrt{\frac{1}{n}}=0$; $\lim_{n\to\infty}\sqrt{\frac{n}{n}}=1$

$f:[0,1]\to\mathbb{R}$ $\to$ continuă $\Rightarrow$ integrabilă



$\|\Delta_n\| = \frac{1}{n} \xrightarrow{n\to\infty} 0$ $\qquad \xi_i \in \left[\frac{i-1}{n}, \frac{i}{n}\right]$; $\xi_i = \frac{i}{n}$

$a_n = \sigma_{\Delta_n}(f, \xi_i) \Rightarrow$

$$\lim_{n\to\infty} a_n = \lim_{\|\Delta_n\|\to 0} \sigma_{\Delta_n}(f,\xi_i) = \int_0^1 f(x)\,dx = \int_0^1 \sqrt{x}\cdot dx$$

$$= \int_0^1 x^{\frac{1}{2}}\cdot dx = \frac{x^{\frac{3}{2}}}{\frac{3}{2}}\Big|_0^1 = \frac{2}{3}(1-0) = \frac{2}{3}$$

$$\boxed{\Rightarrow \lim_{n\to\infty} x_n = \frac{2}{3}}$$

$\lim\limits_{n\to\infty} \sqrt[n]{\frac{(n!)^2}{(2n)!\cdot 8^n}}$ Criteriul Cauchy - d'Alembert.

Dacă $x_n > 0$ și $(\exists) \lim\limits_{n\to\infty} \frac{x_{n+1}}{x_n} = l \Rightarrow (\exists) \lim \sqrt[n]{x_n} = l$

$x_n = \frac{(n!)^2}{(2n)!\cdot 8^n} > 0$; $\lim\limits_{n\to\infty} \frac{x_{n+1}}{x_n} = \lim \frac{((n+1)!)^2}{(2n+2)!\cdot 8^{n+1}} \cdot \frac{(2n)!\cdot 8^n}{(n!)^2} =$

$= \lim\limits_{n\to\infty} \frac{\cancel{(n!)^2}(n+1)^2}{\cancel{(2n)!}(2n+1)(2n+2)\cdot \cancel{8^{n+1}}\,8} \cdot \frac{\cancel{(2n)!}\cdot \cancel{8^n}}{\cancel{(n!)^2}} =$

$= \lim\limits_{n\to\infty} \frac{(n+1)^2}{8(2n+1)\,2(n+1)} = \lim\limits_{n\to\infty} \frac{n+1}{16(2n+1)} \overset{\frac{\infty}{\infty}}{=} \frac{1}{16} \cdot \lim\limits_{n\to\infty} \frac{n(1+\frac{1}{n})}{n(2+\frac{1}{n})}$

$= \frac{1}{16} \cdot \frac{1}{2} = \frac{1}{32} \Rightarrow \lim\limits_{n\to\infty} \sqrt[n]{\frac{n!}{(2n)!\cdot 8^n}} = \frac{1}{32}$

<u>Criteriul raportului.</u>

Dacă $x_n > 0$ și $(\exists) \lim\limits_{n\to\infty} \frac{x_{n+1}}{x_n} = l$ și $l \in [0,1) \Rightarrow$

$\Rightarrow \lim\limits_{n\to\infty} x_n = 0.$

$\lim\limits_{n\to\infty} \frac{x_{n+1}}{x_n} = \frac{1}{32} < 1 \Rightarrow \lim\limits_{n\to\infty} x_n = \lim \frac{n!}{(2n)!\cdot 8^n} = 0.$

$\lim\limits_{n\to\infty} \sqrt[n]{x_n} = \lim (x_n)^{\frac{1}{n}} = (0^0) = \frac{1}{32} < 1$

Seria $\sum\limits_{n=1}^{\infty} x_n \to$ convergentă

(4) $\sum\limits_{n=1}^{\infty} \frac{(-1)^{n-1}}{n^\alpha}$; $\alpha > 0$

$\alpha > 1 \Rightarrow \left|\frac{(-1)^{n-1}}{n^\alpha}\right| = \frac{1}{n^\alpha}$; $\sum \frac{1}{n^\alpha}$ conv $\Rightarrow \sum\limits_1^\infty \frac{(-1)^{n-1}}{n^\alpha}$ AC deci și C

$\alpha = 1 \to \sum\limits_1^\infty \frac{(-1)^{n-1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{(-1)^{n-1}}{n} + \dots$

$\alpha_n = \frac{1}{n} \searrow 0 \overset{\text{Leibniz}}{\Rightarrow} \Rightarrow \sum \frac{(-1)^{n-1}}{n}$ conv

$\alpha \in (0,1) \Rightarrow$ Seria nu: $\sum \frac{1}{n^\alpha} \to$ div pt. $\alpha \in (0,1)$

$\sum (-1)^{n-1} \cdot \frac{1}{n^\alpha}$; $\alpha_n = \frac{1}{n^\alpha} \searrow 0 \Rightarrow \to$ conv (Leibniz)

semiconvergentă

Serii de puteri An I D 13.12.2020

TA suplim.

- calculul razei, intervalului și mulțimii de convergență pt o serie de puteri:

$$\sum_{n=0}^{\infty} a_n \cdot x^n \; ; \; x \in \mathbb{R} \; ; \; \rho = \lim_{n\to\infty} \frac{1}{\sqrt[n]{|a_n|}} = \lim_{n\to\infty} \left|\frac{a_n}{a_{n+1}}\right|$$

$$\sum_{n=1}^{\infty} \frac{x^n}{n(n+1)} \; | \; a_n = \frac{1}{n(n+1)} > 0$$

$$\rho = \lim_{n\to\infty} \left|\frac{a_n}{a_{n+1}}\right| = \lim_{n\to\infty} \frac{\frac{1}{n(n+1)}}{\frac{1}{(n+1)(n+2)}} = \lim_{n\to\infty} \frac{n+2}{n} = \lim_{\infty} \frac{n(1+\frac{2}{n})}{n} = 1$$

$\Rightarrow I = (-1, 1)$

$x = 1 \Rightarrow \sum_{n=1}^{\infty} \frac{1}{n(n+1)} \; ; \; \sum_{1}^{\infty} \frac{1}{n^2} \to$ conv

$\lim_{n\to\infty} \frac{\frac{1}{n(n+1)}}{\frac{1}{n^2}} = 1 \in (0, +\infty) \Rightarrow$ seriile au aceeași natură

$\sum_{1}^{\infty} \frac{1}{n^2}$ conv (seria armonică $\alpha = 2$ $\sum_{1}^{\infty} \frac{1}{n^\alpha}$; $\alpha > 1$ conv)

$x = -1 \Rightarrow \sum_{1}^{\infty} \frac{(-1)^{n-1}}{n(n+1)} \to$ serie alternată

seria modulelor $\longrightarrow \sum_{1}^{\infty} \frac{1}{n(n+1)} \to$ conv. $\Rightarrow$

$x = -1$ se obține o serie AC, deci nu conv.

$\Rightarrow A = [-1, 1]$. $\Rightarrow S(x) = \sum_{1}^{\infty} \frac{x^n}{n(n+1)}$

uniform convergentă $[-1, 1]$

Ex. $1 + x + x^2 + \dots + x^n + \dots \quad x \in \mathbb{R}$

$$S_n(x) = 1 + x + x^2 + \dots + x^n = \frac{1 - x^{n+1}}{1 - x}$$

$\rho = \lim_{n\to\infty} \frac{1}{1} = 1$; $a_n = 1 \Rightarrow I = (-1, 1)$

$x = 1 \to$ div $\quad S(1) = 1 + 1 + 1 + 1 + 1 + \dots = +\infty$

$x = -1 \to$ div $\quad S(-1) = 1 - 1 + 1 - 1 + 1 - 1 \dots \to$ div

$\to A = (-1, 1)$

$$S(x) = \lim_{n\to\infty} \frac{1 - x^{n+1}}{1 - x} = \lim_{n\to\infty} \frac{1}{1 - x} \quad |x| < 1 \Rightarrow \lim_{n\to\infty} x^{n+1} = 0$$

$$1 + x + x^2 + \dots + x^n + \dots = \frac{1}{1 - x} \; , \; (\forall) \; x \in (-1, 1)$$

$$\sum_{n=0}^{\infty} x^n = 1+x+x^2+\dots+x^n+\dots = \frac{1}{1-x} \; ; \; x\in(-1,1)$$

$$x \to -x \Rightarrow \sum_{n=0}^{\infty} (-1)^n \cdot x^n = 1-x+x^2-x^3+\dots+(-1)^n x^n+\dots$$

$$a_n = (-1)^n \; ; \; \lim_{n\to\infty} \left|\frac{a_n}{a_{n+1}}\right| = \lim_{n\to\infty}\left|\frac{(-1)^n}{(-1)^{n+1}}\right| = 1 \; ; \; \rho = 1$$
$$x\in(-1,1)$$

$$\Rightarrow \sum_{n=1}^{\infty}(-1)^n \cdot x^n = \frac{1}{1+x} \; ; \; x\in(-1,1)$$

$$\sum_{n=1}^{\infty} a_n x^n = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots + \frac{(-1)^{n-1}\cdot x^n}{n} + \dots = \sum_{n=1}^{\infty}(-1)^{n-1}\cdot\frac{x^n}{n}$$

$$a_n = \frac{(-1)^{n-1}}{n} \; ; \; \rho = \lim_{n\to\infty}\left|\frac{(-1)^{n-1}}{n}\cdot\frac{n+1}{(-1)^n}\right| = \lim\left|\frac{a_n}{a_{n+1}}\right| =$$

$$= \lim_{\infty}\frac{n+1}{n} = 1. \Rightarrow \rho = 1 \Rightarrow I = (-1,1)$$

$$x=-1 \Rightarrow \sum_{n=1}^{\infty}(-1)^{n-1}\cdot\frac{(-1)^n}{n} = \sum_{1}^{\infty}\frac{(-1)^{2n-1}}{n} = -\sum_{1}^{\infty}\frac{1}{n}$$

$\sum_{n=1}^{\infty}\frac{1}{n} \to$ seria armonică – divergentă.

$$x=1 \Rightarrow \sum_{n=1}^{\infty}(-1)^{n-1}\cdot\frac{1}{n} = 1-\frac{1}{2}+\frac{1}{3}-\frac{1}{4}+\dots+\frac{(-1)^{n-1}}{n}+\dots$$

$\to$ seria armonică alternată, care este cu semne contrare, conform criteriului lui Leibniz

seria modulelor este divergentă, fiind egală cu $\sum_{n=1}^{\infty}\frac{1}{n} \to$ divergentă.

$$\Rightarrow \sum_{n=1}^{\infty}\frac{(-1)^{n-1}}{n} \to \text{semiconvergentă.}$$

$\Rightarrow$ mulțimea de convergență este $A=(-1,1]$.

<u>Teorema a II-a a lui Abel</u>. $\sum_{n=0}^{\infty} a_n\cdot x^n$; ρ = raza de convergență. Dacă seria este convergentă în $x=\rho$ (sau în $x=-\rho$), atunci suma sa $S(x): (-\rho,\rho)\to\mathbb{R}$ este o funcție continuă la stânga în $x=\rho$ (este o funcție continuă la dreapta în $x=-\rho$).

$$x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots + \frac{(-1)^{n-1} \cdot x^n}{n} + \dots = \ln(1+x)$$

$x \in (-1, 1]$; $\ln(1+x) : (-1, +1) \to \mathbb{R}$.

Seria este cont. în $x = 1 \Rightarrow \lim_{\substack{x \to 1 \\ x < 1}} S(x) = S(1) = \ln 2$

$$\Leftrightarrow 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{(-1)^{n-1}}{n} + \dots = \ln 2$$

Derivate parțiale

Fie $u : \mathbb{R}^2 \to \mathbb{R}$; $u(x,y) = -y^2 + e^{x^3 - y^2}$. Să se calculeze derivatele parțiale de ordinul 1 ale lui u și să se arate că aceasta verifică ecuația:

$$2y \cdot \frac{\partial u}{\partial x} + 3x^2 \cdot \frac{\partial u}{\partial y} + 6x^2 y = 0, \ (\forall)(x,y) \in \mathbb{R}^2.$$

(2) $\frac{\partial u}{\partial x} = u'_x = 0 + e^{x^3 - y^2} \cdot (3x^2 - 0) = 3x^2 \cdot e^{x^3 - y^2}$; $(e^{u(x)})' = e^{u(x)} \cdot u'(x)$

$\frac{\partial u}{\partial y} = -2y + e^{x^3 - y^2} \cdot (0 - 2y) = -2y - 2y \cdot e^{x^3 - y^2} \quad | \cdot 3x^2$

$2y \cdot \frac{\partial u}{\partial x} + 3x^2 \cdot \frac{\partial u}{\partial y} + 6x^2 y = \cancel{6x^2 y \cdot e^{x^3 - y^2}} - \cancel{6x^2 y} - \cancel{6x^2 y \cdot e^{x^3 - y^2}} + \cancel{6x^2 y}$

$= 0$

(5) $u(x,y) = \ln(x^3 + y^2)$; $\frac{\partial u}{\partial x} = ?$ $\frac{\partial u}{\partial y} = ?$

$$\boxed{(\ln(u(x)))' = \frac{u'(x)}{u(x)}}$$

$(2y+1) \frac{\partial u}{\partial x} - 3x^2 \cdot \frac{\partial u}{\partial y} - \frac{3x^2}{x^3 + y^2} = 0.$

$\frac{\partial u}{\partial x} = \frac{(x^3 + y^2)'_x}{x^3 + y^2} = \frac{3x^2}{x^3 + y^2}$; $\frac{\partial u}{\partial y} = \frac{(x^3 + y^2)'_y}{x^3 + y^2} = \frac{2y}{x^3 + y^2}$

$\Rightarrow (2y+1) \cdot \frac{3x^2}{x^3 + y^2} - 3x^2 \cdot \frac{2y}{x^3 + y^2} - \frac{3x^2}{x^3 + y^2} = \frac{\cancel{6x^2 y} + 3x^2 - \cancel{6x^2 y} - 3x^2}{x^3 + y^2} = 0.$

(13) $w(x,y) = (x^2 + y^2) \cdot \sin\left(\frac{y}{x}\right)$; $x \neq 0$

$x \cdot \frac{\partial w}{\partial x} + y \frac{\partial w}{\partial y} = 2 \cdot w(x,y)$

$(f \cdot g)' = f' \cdot g + f \cdot g'$

$(\sin u(x))' = \cos u \cdot u'$

$\left(\frac{1}{x}\right)' = -\frac{1}{x^2}$

$(x^n)' = n \cdot x^{n-1}$

$x \cdot \frac{\partial w}{\partial x} = 2x \cdot \sin\left(\frac{y}{x}\right) + (x^2 + y^2) \cdot \cos\left(\frac{y}{x}\right) \cdot \left(-\frac{y}{x^2}\right)$

$y \cdot \frac{\partial w}{\partial y} = 2y \cdot \sin\left(\frac{y}{x}\right) + (x^2 + y^2) \cdot \cos\left(\frac{y}{x}\right) \cdot \left(\frac{1}{x}\right)$

$2x^2 \cdot \sin\frac{y}{x} - \cancel{\frac{y}{x}(x^2 + y^2) \cdot \cos\frac{y}{x}} + 2y^2 \cdot \sin\frac{y}{x} + \cancel{\frac{y}{x} \cdot (x^2 + y^2) \cdot \cos\frac{y}{x}} = 2w(x,y)$

Să se calculeze diferențiala de ordinul 1 și 2 a funcției: $f(x,y) = 27x^3 - 54x^2y + 36xy^2 - 8y^3$ în punctul $M(1,1)$.

$$df = \left(\frac{\partial f}{\partial x}\cdot dx + \frac{\partial f}{\partial y}\cdot dy\right)(1,1) = 9\cdot dx - 6\,dy$$

$$\frac{\partial f}{\partial x} = 81x^2 - 108xy + 36y^2 \;;\; \frac{\partial f}{\partial y} = -54x^2 + 72xy - 24y^2$$

$$d^2f = \frac{\partial^2 f}{\partial x^2}\cdot(dx)^2 + 2\cdot\frac{\partial^2 f}{\partial x\partial y}\cdot dx\,dy + \frac{\partial^2 f}{\partial y^2}\cdot(dy)^2$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = 162x - 108y \qquad \frac{\partial^2 f}{\partial x^2}(1,1) = 54$$

$$\frac{\partial^2 f}{\partial x\partial y} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = -108x + 72y \qquad \frac{\partial^2 f}{\partial x\partial y}(1,1) = -36$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = 72x - 48y \qquad \frac{\partial^2 f}{\partial y^2}(1,1) = +24$$

$$\frac{\partial^2 f}{\partial y\partial x} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = -108x + 72y$$

$$d^2f(1,1) = 54\cdot dx^2 + 2\cdot(-36)\,dx\,dy + 24\cdot dy^2$$
$$= 54\,dx^2 - 72\cdot dx\,dy + 24\,dy^2$$

$$\underbrace{(x^2+2xy-y^2)}_{P(x,y)}\cdot dx + \underbrace{(x^2-2xy-y^2)}_{Q(x,y)}\cdot dy = du$$

$$\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$$

$$P(x,y) = x^2+2xy-y^2$$
$$Q(x,y) = x^2-2xy-y^2$$

$$\frac{\partial P}{\partial y} = 2x - 2y$$
$$\parallel$$
$$\frac{\partial Q}{\partial x} = 2x - 2y$$

$$\begin{cases} \underbrace{\dfrac{\partial u}{\partial x}}_{P}\cdot dx + \underbrace{\dfrac{\partial u}{\partial y}}_{Q}\cdot dy = du \\ \dfrac{\partial P}{\partial y} = \dfrac{\partial}{\partial y}\left(\dfrac{\partial u}{\partial x}\right) = \dfrac{\partial^2 u}{\partial x\partial y} \\ \dfrac{\partial Q}{\partial x} = \dfrac{\partial}{\partial x}\left(\dfrac{\partial u}{\partial y}\right) = \dfrac{\partial^2 u}{\partial y\partial x} \end{cases}$$

11.12.2020

Prezentare Probele de evaluare An I și + An II

(1°) Fie funcția $f(x,y) = x^3 + 2x^2y - 3xy^2 + 4y^2 - 5xy + 6x - 4y + 3$.

$d^2f(1,-1)$.

$$d^2f(x,y) = \left(\frac{\partial}{\partial x}\cdot dx + \frac{\partial}{\partial y}\cdot dy\right)^{(2)} f(x,y) =$$

$$= \frac{\partial^2 f(x,y)}{\partial x^2} dx^2 + 2\frac{\partial^2 f(x,y)}{\partial x \partial y} dx\,dy + \frac{\partial^2 f(x,y)}{\partial y^2} dy^2$$

$$\frac{\partial^2 f}{\partial x^2}(1,-1) \; ; \; \frac{\partial^2 f}{\partial x \partial y}(1,-1) \; ; \; \frac{\partial^2 f}{\partial y^2}(1,-1)$$

$$\frac{\partial f}{\partial x}(x,y) = 3x^2 + 4xy - 3y^2 - 5y + 6$$

$$\frac{\partial^2 f}{\partial x^2}(x,y) = 6x + 4y$$

$$\frac{\partial^2 f(x,y)}{\partial x \partial y} = \frac{\partial}{\partial y}\left(\frac{\partial f(x,y)}{\partial x}\right) = \underline{4x - 6y - 5}$$

$$\frac{\partial f(x,y)}{\partial y} = 2x^2 - 6xy + 8y - 5x - 4$$

$$\frac{\partial^2 f}{\partial y^2} = -6x + 8$$

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \underline{4x - 6y - 5}$$

$$\frac{\partial^2 f}{\partial x^2}(1,-1) = 6 - 4 = 2$$

$$\frac{\partial^2 f}{\partial x \partial y}(1,-1) = 4 + 6 - 5 = 5$$

$$\frac{\partial^2 f}{\partial y^2}(1,-1) = -6 + 8 = 2$$

$$d^2f(1,-1) = 2\cdot(dx)^2 + 2\cdot 5\, dx\,dy + 2\cdot(dy)^2$$

(2°) Să se determine extremele funcției:

$f(x,y) = xy(a-x-y)$, $a>0$.

$f(x,y) = axy - x^2y - xy^2$

Det. punctelor staționare: (cond. necesare pt extrem)

$$\begin{cases} \frac{\partial f}{\partial x} = 0 \\ \frac{\partial f}{\partial y} = 0 \end{cases} ; \begin{cases} ay - 2xy - y^2 = 0 \\ ax - x^2 - 2xy = 0 \end{cases} ; \begin{cases} y(a-2x-y) = 0 \\ x(a-x-2y) = 0 \end{cases}$$

$\Rightarrow x=0, y=0 \rightarrow M_1(0,0)$ = punct staționar

$$\begin{cases} a-2x-y=0 \\ a-x-2y=0 \end{cases} ; \begin{cases} 2x+y=a \\ x+2y=a \quad / -2 \end{cases} \Rightarrow \begin{array}{l} 2x+y=a \\ -2x-4y=-2a \\ \hline / \ -3y=-a \end{array} \boxed{y=\frac{a}{3}}$$

$x = a-2y$; $x = a - \frac{2a}{3} = \boxed{\frac{a}{3} = x}$

$M_2\left(\frac{a}{3}, \frac{a}{3}\right)$.

condiția suficientă pt extrem

$$H = \begin{pmatrix} \frac{\partial^2 f}{\partial x^2} & \frac{\partial^2 f}{\partial x \partial y} \\ \frac{\partial^2 f}{\partial y \partial x} & \frac{\partial^2 f}{\partial y^2} \end{pmatrix} \rightarrow \text{în fiecare punct staționar.}$$

$$\begin{cases} \frac{\partial^2 f}{\partial x^2} = \frac{\partial}{\partial x}(ay - 2xy - y^2) = -2y \\ \frac{\partial^2 f}{\partial x \partial y} = \frac{\partial}{\partial y}(ay - 2xy - y^2) = a - 2x - 2y \\ \frac{\partial^2 f}{\partial y^2} = \frac{\partial}{\partial y}(ax - x^2 - 2xy) = -2x \end{cases}$$

$H(0,0) = \begin{pmatrix} 0 & a \\ a & 0 \end{pmatrix} \Rightarrow M_1(0,0)$ nu este punct de extrem

$H\left(\frac{a}{3}, \frac{a}{3}\right) = \begin{pmatrix} -2\frac{a}{3} & -\frac{a}{3} \\ -\frac{a}{3} & -\frac{2a}{3} \end{pmatrix}$ $\Delta_1 = -\frac{2a}{3} < 0$

$\Delta_2 = \det H = \frac{4a^2}{9} + \frac{a^2}{9} = \frac{5a^2}{9}$

$d^2f\left(\frac{a}{3}, \frac{a}{3}\right) = < 0$

$a - 2a - \frac{2a}{3} = a - 4a = -a$

$$\sum_{n=2}^{\infty} \frac{\ln(n+1) - \ln n}{\ln(n+1) \cdot \ln n}$$

$$u_n = \frac{\ln(n+1) - \ln n}{\ln(n+1) \cdot \ln n} = \frac{\ln(n+1)}{\ln(n+1) \cdot \ln n} - \frac{\ln n}{\ln(n+1) \cdot \ln n} =$$

$$u_n = \frac{1}{\ln n} - \frac{1}{\ln(n+1)}$$

$n=2: \quad u_2 = \frac{1}{\ln 2} - \cancel{\frac{1}{\ln 3}}$

$n=3: \quad u_3 = \cancel{\frac{1}{\ln 3}} - \cancel{\frac{1}{\ln 4}}$

$\dots$

$n-1: \quad u_{n-1} = \cancel{\frac{1}{\ln(n-1)}} - \cancel{\frac{1}{\ln n}}$

$n: \quad u_n = \cancel{\frac{1}{\ln n}} - \frac{1}{\ln(n+1)}$

$$S_n = u_2 + u_3 + \dots + u_n = \frac{1}{\ln 2} - \frac{1}{\ln(n+1)}$$

$$S = \lim_{n \to \infty} S_n = \lim_{n \to \infty} \left(\frac{1}{\ln 2} - \frac{1}{\ln(n+1)} \right) = \frac{1}{\ln 2}$$

Fie seria de numere:

$$\sum_{n=1}^{\infty} \frac{(-1)^n}{n(n+1) \cdot 2^{2n}} \rightarrow \text{funcţie serie!}$$

$$|u_n| = \frac{1}{n(n+1) \cdot 4^n} \; ; \quad \lim \frac{|u_{n+1}|}{u_n} = \lim \frac{\frac{1}{(n+1)(n+2) \cdot 4^{n+1}}}{\frac{1}{n(n+1) \cdot 4^n}} =$$

$$= \lim_{n \to \infty} \frac{n}{(n+2)4} = \frac{1}{4} \cdot \lim \frac{n}{n(1+\frac{2}{n})} = \frac{1}{4} < 1$$

Seria modulelor este cauzată $\Rightarrow$ seria este A.C.

Pt calculul sumei:

Asociem seria de puteri $\sum_{n=1}^{\infty} \frac{x^n}{n(n+1)} = \sum_{1}^{\infty} a_n \cdot x^n$

$$\rho = \lim_{n \to \infty} \left| \frac{a_n}{a_{n+1}} \right| = \lim \frac{\frac{1}{n(n+1)}}{\frac{1}{(n+1)(n+2)}} = \lim_{n \to \infty} \frac{n+2}{n} = 1$$

$I = (-1, 1)$; $\quad x = -\frac{1}{4} \Rightarrow \sum^{\infty} \frac{\left(-\frac{1}{4}\right)^n}{n(n+1)} = \sum \frac{(-1)^n}{n(n+1) \cdot 4^n}$

$f(x,y) = (x^2+y^2)\cdot \sin\left(\frac{y}{x}\right)$ Care este ecuația cu derivate parțiale satisfăcută de această funcție?

a) $x\cdot\frac{\partial f}{\partial x} + y\cdot\frac{\partial f}{\partial y} = 2(x^2+y^2)\cdot \sin\frac{y}{x}$

b) $x^2\frac{\partial f}{\partial x} + y^2\frac{\partial f}{\partial y} =$

c)

d)

$$\frac{\partial f}{\partial x} = 2x\cdot\sin\left(\frac{y}{x}\right) + (x^2+y^2)\cdot\cos\left(\frac{y}{x}\right)\cdot\left(-\frac{y}{x^2}\right) \quad |\cdot x$$

$$\frac{\partial f}{\partial y} = 2y\cdot\sin\frac{y}{x} + (x^2+y^2)\cdot\cos\left(\frac{y}{x}\right)\cdot\frac{1}{x} \quad |\cdot y$$

$$x\cdot\frac{\partial f}{\partial x} + y\cdot\frac{\partial f}{\partial y} = 2x^2\cdot\sin\frac{y}{x} + 2y^2\cdot\sin\left(\frac{y}{x}\right) \cancel{- \frac{y}{x}\cdot(x^2+y^2)\cdot\cos\frac{y}{x} + \frac{y}{x}(x^2+y^2)\cdot\cos\frac{y}{x}}$$

$$x\cdot\frac{\partial f}{\partial x} + y\cdot\frac{\partial f}{\partial y} = 2(x^2+y^2)\cdot\sin\frac{y}{x}$$